

Case Study 2 Part 1

Data Overview

- Data extracted from the Federal Reserve Economic Data (FRED)
- Selected series:
 - **INDPRO**: Industrial Production (Total Index)
 - **CPIAUCSL**: Consumer Price Index (All Urban Consumers)
 - **FEDFUNDS**: Effective Federal Funds Rate
- Original sample: January 1959 – December 2019
 - **Effective sample: March 1959 – December 2019** (...Why?)

Economic Interpretation

- The three series capture distinct dimensions of aggregate economic activity
- Following *Bernanke (2005)*, they can be interpreted as:
 - *INDPRO* – Industrial Production → **Real activity / Output**
 - *CPIAUCSL* – Consumer Price Index → **Prices / Inflation**
 - *FEDFUNDS* – Effective Federal Funds Rate → **Monetary policy**

Benchmark Transformations

- *McCracken and Ng (2016)* provide benchmark transformations intended to obtain stationary representations of the FRED-MD series:

Series	Transformation	Interpretation
<i>INDPRO</i>	$\Delta \log(\text{INDPRO}_t)$	Output growth
<i>CPIAUCSL</i>	$\Delta^2 \log(\text{CPIAUCSL}_t)$	Change in inflation
<i>FEDFUNDS</i>	$\Delta \text{FEDFUNDS}_t$	Change in the policy rate

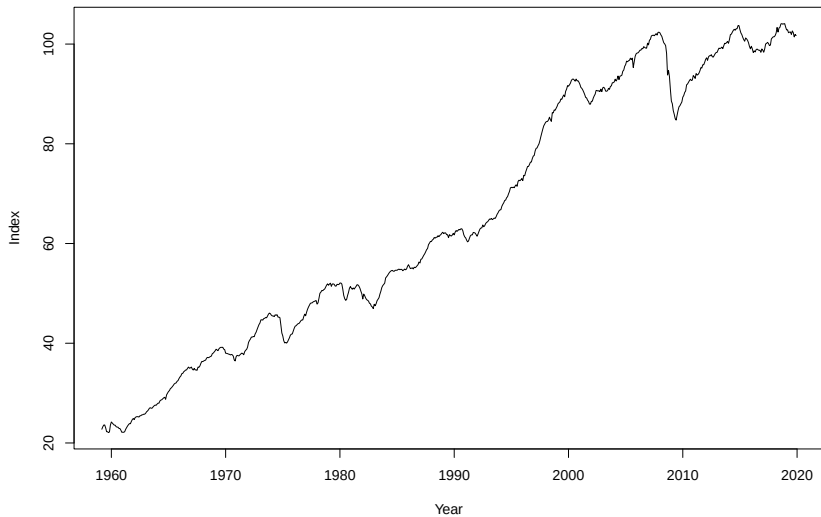
- These transformations are treated as benchmarks rather than assumptions

Series in Levels

- Before transforming the data, we inspect the original series over time

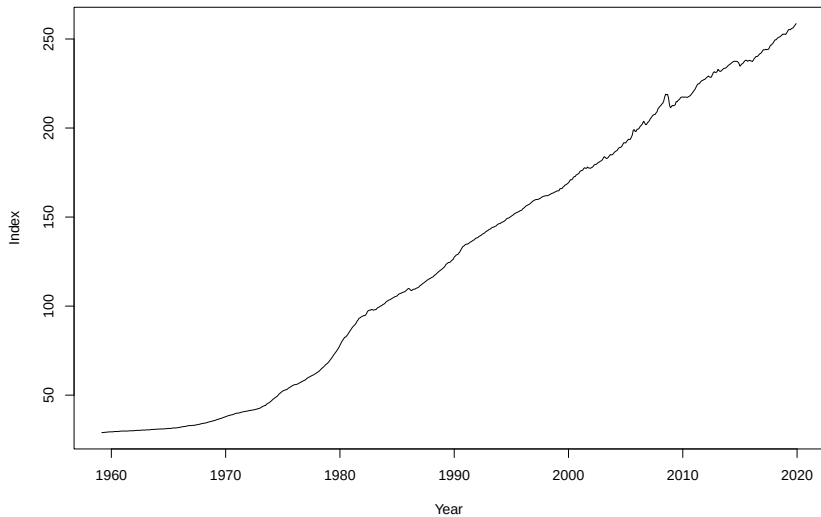
Series in Levels

Industrial Production



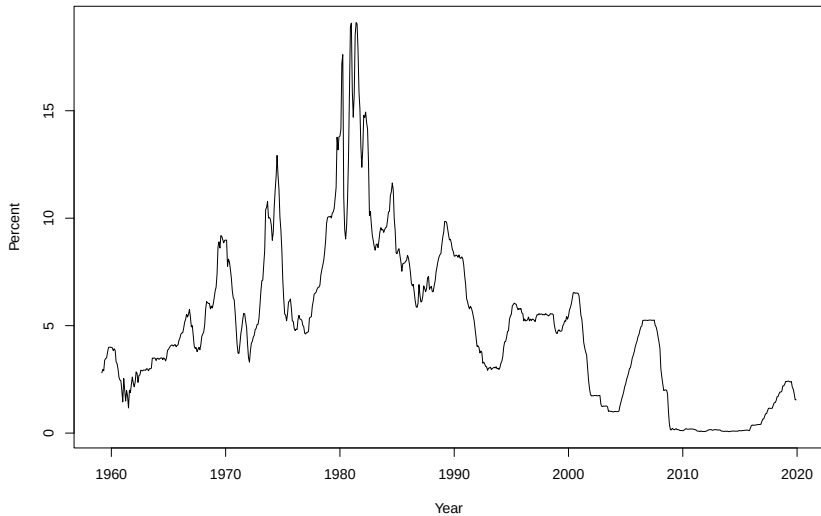
Series in Levels

Consumer Price Index



Series in Levels

Federal Funds Rate



Series in Levels

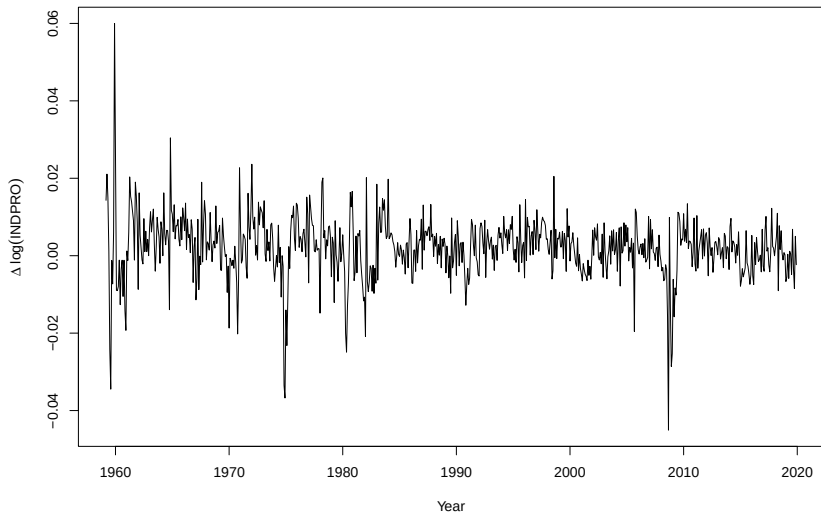
- Before transforming the data, we inspect the original series over time
 - *INDPRO* exhibits a pronounced long-run upward trend and cyclical fluctuations
 - *CPIAUCSL* displays a strong and persistent upward trend
 - *FEDFUNDS* shows persistent movements and substantial variation
- The visual evidence suggests that stationarity cannot be assumed in levels

Transformed Series

- Now, we apply the FRED-MD benchmark transformations:
 - $\Delta \log(INDPRO_t)$
 - $\Delta^2 \log(CPIAUCSL_t)$
 - $\Delta FEDFUNDS_t$

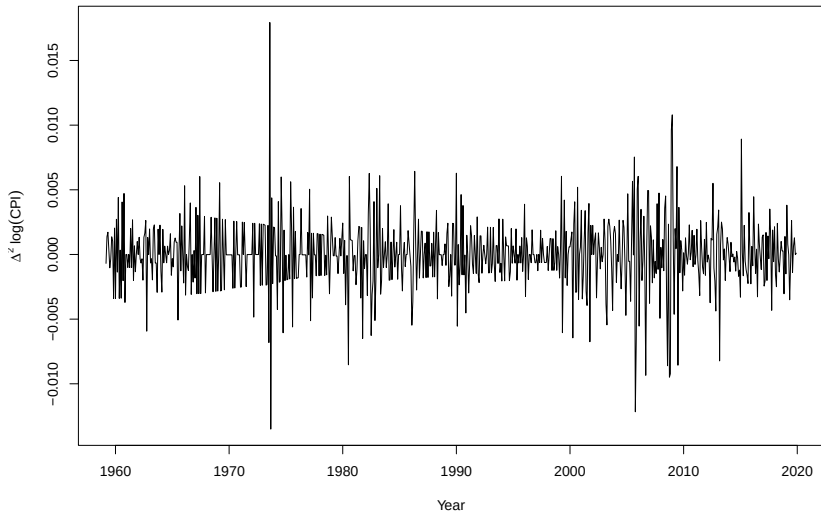
Transformed Series

Output Growth

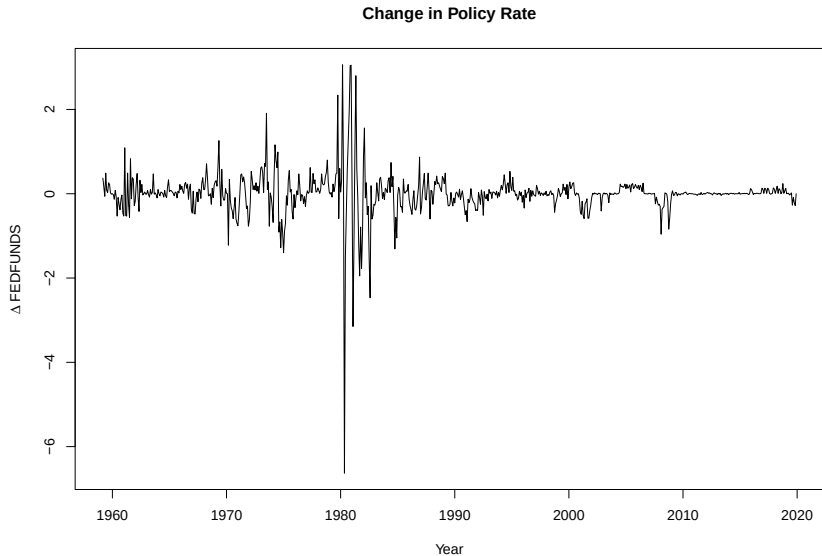


Transformed Series

Change in Inflation



Transformed Series



Transformed Series

- Now, we apply the FRED-MD benchmark transformations:
 - $\Delta \log(INDPRO_t)$
 - $\Delta^2 \log(CPIAUCSL_t)$
 - $\Delta FEDFUNDS_t$
 - The transformations substantially changes the persistence of the series
 - Less persistence and no obvious deterministic trend
 - Consistent with stationarity, but visual inspection is not sufficient to establish it
- ⇒ Still need to formally assess whether the transformed series are stationary

VAR Specification

- Define the vector of stationary variables as

$$\mathbf{y}_t = \begin{bmatrix} \Delta \log(INDPRO_t) \\ \Delta^2 \log(CPIAUCSL_t) \\ \Delta FEDFUNDS_t \end{bmatrix} = \begin{bmatrix} IP_t \\ CPI_t \\ FFR_t \end{bmatrix}$$

- Following *Pfaff (2008)*, specify the reduced-form VAR(p) as

$$\mathbf{y}_t = \mathbf{c} + A_1 \mathbf{y}_{t-1} + \cdots + A_p \mathbf{y}_{t-p} + \mathbf{u}_t$$

Lag Order Selection

- Common methods to determine the lag order given by *Kilian and Lütkepohl (2017)* include:
 - **Top-down testing:** Start with many lags and remove the highest lag step by step if it adds little information
 - ▶ *Adv.:* Simple and directly tests lag relevance
 - ▶ *Lim.:* Depends on the starting order
 - **Bottom-up testing:** Start with few lags and add more only if they meaningfully improve the model
 - ▶ *Adv.:* Favors parsimonious models
 - ▶ *Lim.:* May stop too early and miss longer-run dynamics
 - **Forecast-based selection:** Choose the lag order with the best predictive performance
 - ▶ *Adv.:* Directly targets forecasting
 - ▶ *Lim.:* Depends on horizon and evaluation sample

Information Criteria

- An alternative is to compare candidate VAR specifications using information criteria
 - **AIC** – Comparatively weaker penalty for additional parameters
 - **HQ** – Intermediate penalty for model complexity
 - **SC/BIC** – Stronger penalty, generally favoring more parsimonious models
 - **FPE** – Evaluates the estimated prediction error of competing specifications
- ▶ *Adv.:* Systematic comparison of fit and model complexity
- ▶ *Lim.:* Different criteria may select different lag orders

Lag Selection Results – Information Criteria

- Lag orders are selected using `VARselect()` from the `vars` package, comparing AIC, HQ, SC/BIC and FPE:

Criterion	Selected lag
AIC	14
HQ	4
SC / BIC	3
FPE	14

- AIC selects $p = 14$, while SC/BIC selects $p = 3$
⇒ These define the two candidate lag specifications

VAR Estimation

- Each candidate specification is estimated separately using `VAR()` from the `vars` package
 - AIC specification: `VAR(14)`
 - SC/BIC specification: `VAR(3)`
- For each specification,

$$\hat{\mathbf{y}}_t = \hat{\mathbf{c}} + \hat{A}_1 \mathbf{y}_{t-1} + \cdots + \hat{A}_p \mathbf{y}_{t-p}$$

→ Each matrix $\hat{A}_j$ contains the estimated own- and cross-lag effects among output growth, changes in inflation, and changes in the policy rate.

Coefficient Interpretation

- Each row of $\hat{A}_j$ corresponds to one VAR equation, while each column measures the effect of a lagged system variable
 - Diagonal coefficients capture own-lag persistence
 - Off-diagonal coefficients capture cross-variable predictive effects
 - Coefficients are interpreted conditional on the remaining variables and included lags
- The coefficients describe reduced-form dynamic relationships, not structural causal effects

VAR Stability

- Stability assessed using `roots()` from the `vars` package
- A VAR is stable if all characteristic roots satisfy $|\lambda_i| < 1$

Specification	Maximum root modulus
AIC	0.939
SC / BIC	0.665

- Both specifications satisfy the stability condition
 - AIC: maximum modulus = 0.939 – dynamics are relatively persistent
 - SC/BIC: maximum modulus = 0.665 – shocks decay more rapidly through the system

Residual Diagnostics

- Residual properties are assessed using `serial.test()`, `arch.test()` and `normality.test()` from the `vars` package (*Pfaff, 2008*)

- **Serial correlation:** H_0 – no residual serial correlation

- ▶ AIC VAR(14): $p = 0.0060 \rightarrow \text{reject } H_0$

- ▶ SC/BIC VAR(3): $p < 0.0001 \rightarrow \text{reject } H_0$

Residual dependence remains under both lag specifications

- **ARCH effects:** H_0 – no conditional heteroskedasticity

- ▶ Both specifications: $p < 0.0001 \rightarrow \text{reject } H_0$

Residual variance is not constant over time

- **(Normality:** H_0 – multivariate normality)

- ▶ Both specifications: $p < 0.0001 \rightarrow \text{reject } H_0$

Normality is not required for VAR estimation; its rejection motivates bootstrap-based inference for the subsequent IRFs

Model Selection

- Neither specification fully eliminates residual serial correlation
- The SC/BIC VAR(3) is more parsimonious, while the AIC VAR(14) allows for richer monthly dynamics
- VAR(14) is retained as the model specification for the subsequent IRF analysis
 - **Results should be interpreted cautiously given the remaining residual misspecification!**

